

Liangliang Zhang

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EDUCATION

Rensselaer Polytechnic Institute, Troy, USA

August 2024 - Now

Ph.D. in Computer Science

Central South University, Changsha, China

September 2022 - June 2024

B.S. in Software Engineering

RESEARCH INTEREST

Large Language Models, GraphRAG, Knowledge Graphs, Graph Neural Networks, Trustworthy AI, Data-Centric AI, Data Condensation.

SELECTED HONORS AND AWARDS

- **KDD 2024 Student Travel Award** June 2024
Awarded for outstanding contributions and participation at the KDD 2024 conference.
- **Tutorial Presentation at KDD 2024** May 2024
Co-presented a tutorial on "Safe Multi-Modal Machine Learning" at the KDD 2024 conference, showcasing advancements and best practices in the field.
- **Academic Year Scholarship, School of Computer Science, Central South University**
Awarded the scholarship **four times** of academic excellence and outstanding performance.

PUBLICATION

CONFERENCE AND JOURNAL PUBLICATIONS

- [1] Zhao, Tianyi*, **Zhang, Liangliang***, Y. Ma, and L. Cheng. "A Survey on Safe Multi-Modal Learning System". In: *In Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining(SIGKDD)* (2024). * contributed equally to this work.
- [2] **Zhang, Liangliang**, M. Chen, X. Hu, and L. Deng. "Graph Convolutional Network and Contrastive Learning Small Nucleolar RNA (snoRNA) Disease Associations (GCLSDA): Predicting snoRNA–Disease Associations via Graph Convolutional Network and Contrastive Learning". In: *International Journal of Molecular Sciences* 24.19 (2023), p. 14429.
- [3] S. Wang, **Zhang, Liangliang**, and H. Fan. "An Enhanced Multi-Constraint Optimization Algorithm for Efficient Network Topology Generation". In: *Mathematics* 11.16 (2023), p. 3456.

PREPRINTS AND SUBMISSIONS

- [4] H. Xu, **Zhang, Liangliang**, Y. Ma, S. Zhou, Z. Zheng, and B. Jiajun. "A Survey on Graph Condensation". In: *arXiv preprint arXiv:2402.02000* (2024).
- [5] **Zhang, Liangliang**, H. Bao, and Y. Ma. "Extending Graph Condensation to Multi-Label Datasets: A Benchmark Study". In: *arXiv preprint arXiv:2412.17961* (2024).

TEACHING & RESEARCH EXPERIENCE

DAMI Lab, | *Research Assistant* January 2025 – Present
Research on Data-Centric AI (Data Augmentation, Active Learning, Data Valuation, Data Condensation) and Large Language Models (GraphRAG, Knowledge Graphs for LLMs).

RPI CSCI 1100: Computer Science, | *Teaching Assistant* August 2024 – December 2024
Graded exams and assignments, proctored tests, led lab sessions, and held office hours to support student learning (20 hrs/week).

Competitive Programming Club | *Team Member* July 2022 – September 2023
Research on Data Mining and Graph Neural Networks and Bioinformatics.

Big Data Computing | *Team Member* September 2021 - September 2022
Researched and implemented swarm intelligence algorithms for resource scheduling.

INTERNSHIP

Institute of Automation, Chinese Academy of Sciences March - September 2023

- Enhanced skills in Linux and Docker for system operations.
- Debugged deep learning interfaces and developed attack algorithms.
- Optimized CW attacks and implemented AutoAttack and CopyCAT methods.

Hunan Wuche Education Technology Co., Ltd. August - September 2022

- Co-designed an online tank battle game using JavaFX and FXGL.
- Led project design and presentations, enhancing Java expertise in UI, SQL, and Javax.mail.

China Hi-Tech Group June - July 2022

- Developed a shopping mall app with Vue and SpringBoot.
- Directed program design, task coordination, and team meetings.
- Gained proficiency in SpringBoot, Vue, and JavaScript.

Guangdong QST Green Soft Training Education Technology Co., Ltd. June - July 2021

- Created an online Gomoku game in C++.
- Implemented user authentication, game modes, and enhanced C++ skills.